Question 3: Matrix Chain Multiplication

Problem Statement

Given five matrices with the following dimensions:

$$M_1 : 1 \times 5, \quad M_2 : 5 \times 2, \quad M_3 : 2 \times 3, \quad M_4 : 3 \times 9, \quad M_5 : 9 \times 4$$

Find the minimum number of scalar multiplications required to compute $\prod_{j=1}^{5} M_j$ and an optimal parenthesization.

Dynamic Programming Formulation

Let $p = [1, 5, 2, 3, 9, 4]$ be the dimension array so that matrix M_i has dimensions $p[i] \times p[i+1]$ (0-indexed). Define $m[i][j]$ as the minimum number of scalar multiplications needed to compute $M_i \cdot M_{i+1} \cdots M_j$. The base case is $m[i][i] = 0$ (a single matrix needs no multiplications), and the recurrence for $i < j$ is:

$$m[i][j] = \min_{i \le k < j} \Big\{ m[i][k] + m[k+1][j] + p[i] \cdot p[k+1] \cdot p[j+1] \Big\}$$

We also store $s[i][j] = k$ (the optimal split point) to reconstruct the parenthesization.

Algorithm

```
def optimal_matrix_mult(matrices):
    N = len(matrices)
    m = [[float('inf')] * N for _ in range(N)]
    s = [[0] * N for _ in range(N)]

    for i in range(N):
        m[i][i] = 0

    for L in range(2, N + 1):          # chain length
        for i in range(N - L + 1):
            j = i + L - 1
            for k in range(i, j):
                cost = (m[i][k] + m[k + 1][j]
                        + matrices[i][0] * matrices[k][1] * matrices[j][1])
                if cost < m[i][j]:
                    m[i][j] = cost
                    s[i][j] = k

    def construct_parens(i, j):
        if i == j:
            return f"M{i + 1}"
        k = s[i][j]
        left = construct_parens(i, k)
        right = construct_parens(k + 1, j)
        return f"({left} {right})"

    return m[0][N - 1], construct_parens(0, N - 1), m, s
```

Computation

We fill the m and s tables by increasing chain length L:

Chain length $L = 2$:

$$\begin{aligned}
m[0][1] &= p[0] \cdot p[1] \cdot p[2] = 1 \cdot 5 \cdot 2 = 10, \quad s[0][1] = 0 \\
m[1][2] &= p[1] \cdot p[2] \cdot p[3] = 5 \cdot 2 \cdot 3 = 30, \quad s[1][2] = 1 \\
m[2][3] &= p[2] \cdot p[3] \cdot p[4] = 2 \cdot 3 \cdot 9 = 54, \quad s[2][3] = 2 \\
m[3][4] &= p[3] \cdot p[4] \cdot p[5] = 3 \cdot 9 \cdot 4 = 108, \quad s[3][4] = 3
\end{aligned}$$

Chain length $L = 3$:

$$\begin{aligned}
m[0][2]: \quad & k = 0 : m[0][0] + m[1][2] + 1 \cdot 5 \cdot 3 = 0 + 30 + 15 = 45 \\
& k = 1 : m[0][1] + m[2][2] + 1 \cdot 2 \cdot 3 = 10 + 0 + 6 = \mathbf{16} \quad \Rightarrow s[0][2] = 1 \\
m[1][3]: \quad & k = 1 : m[1][1] + m[2][3] + 5 \cdot 2 \cdot 9 = 0 + 54 + 90 = \mathbf{144} \\
& k = 2 : m[1][2] + m[3][3] + 5 \cdot 3 \cdot 9 = 30 + 0 + 135 = 165 \quad \Rightarrow s[1][3] = 1 \\
m[2][4]: \quad & k = 2 : m[2][2] + m[3][4] + 2 \cdot 3 \cdot 4 = 0 + 108 + 24 = 132 \\
& k = 3 : m[2][3] + m[4][4] + 2 \cdot 9 \cdot 4 = 54 + 0 + 72 = \mathbf{126} \quad \Rightarrow s[2][4] = 3
\end{aligned}$$

Chain length $L = 4$:

$$\begin{aligned}
m[0][3]: \quad & k = 0 : m[0][0] + m[1][3] + 1 \cdot 5 \cdot 9 = 0 + 144 + 45 = 189 \\
& k = 1 : m[0][1] + m[2][3] + 1 \cdot 2 \cdot 9 = 10 + 54 + 18 = 82 \\
& k = 2 : m[0][2] + m[3][3] + 1 \cdot 3 \cdot 9 = 16 + 0 + 27 = \mathbf{43} \quad \Rightarrow s[0][3] = 2 \\
m[1][4]: \quad & k = 1 : m[1][1] + m[2][4] + 5 \cdot 2 \cdot 4 = 0 + 126 + 40 = \mathbf{166} \\
& k = 2 : m[1][2] + m[3][4] + 5 \cdot 3 \cdot 4 = 30 + 108 + 60 = 198 \\
& k = 3 : m[1][3] + m[4][4] + 5 \cdot 9 \cdot 4 = 144 + 0 + 180 = 324 \quad \Rightarrow s[1][4] = 1
\end{aligned}$$

Chain length $L = 5$:

$$\begin{aligned}
m[0][4]: \quad & k = 0 : m[0][0] + m[1][4] + 1 \cdot 5 \cdot 4 = 0 + 166 + 20 = 186 \\
& k = 1 : m[0][1] + m[2][4] + 1 \cdot 2 \cdot 4 = 10 + 126 + 8 = 144 \\
& k = 2 : m[0][2] + m[3][4] + 1 \cdot 3 \cdot 4 = 16 + 108 + 12 = 136 \\
& k = 3 : m[0][3] + m[4][4] + 1 \cdot 9 \cdot 4 = 43 + 0 + 36 = \mathbf{79} \quad \Rightarrow s[0][4] = 3
\end{aligned}$$

DP Tables

Table m (minimum costs):

$m[i][j]$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 0$	0	10	16	43	**79**
$i = 1$		0	30	144	166
$i = 2$			0	54	126
$i = 3$				0	108
$i = 4$					0

Table s (split points):

$s[i][j]$	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$
$i=0$	–	0	1	2	3
$i=1$		–	1	1	1
$i=2$			–	2	3
$i=3$				–	3
$i=4$					–

Result

$$\text{Minimum scalar multiplications} = m[0][4] = \boxed{79}$$

Optimal Parenthesization: Tracing back through the s table:

- $s[0][4] = 3$: split into $(M_1 \cdots M_4)$ and (M_5)
- $s[0][3] = 2$: split into $(M_1 \cdots M_3)$ and (M_4)
- $s[0][2] = 1$: split into $(M_1 \cdot M_2)$ and (M_3)

$$\textbf{Optimal Parenthesization:} \quad ((((M_1\, M_2)\, M_3)\, M_4)\, M_5)$$

Complexity Analysis

The algorithm has **time complexity** $O(n^3)$ due to three nested loops (chain length, start index, split point), where n is the number of matrices. The **space complexity** is $O(n^2)$ for storing the m and s tables.